**DATA 260 Homework 1 Report**

**GITHUB REPOSITORY: https://github.com/rutujasaste06/data260-5825**

Student ID- **020065825**

SID4- **5825**

PORT_BASE - **8425**

PREFIX - **s5825**

SEED - **5825**

VERIFY_SEED – **265825**

DOMAIN_ID - **1**

**DOMAIN - Clinical trial listings**

HARDWARE **- 13th Gen Intel Core i7-13700HX (2.10 GHz), 16GB RAM, NVIDIA GeForce RTX 4070 Laptop GPU (8GB)**

LOCAL MODEL USED: **qwen3:8b (Ollama)**

TAGGED COMMIT HASH: **714d9fb7436bf5b5e06349ad43ab117b65cc6355**

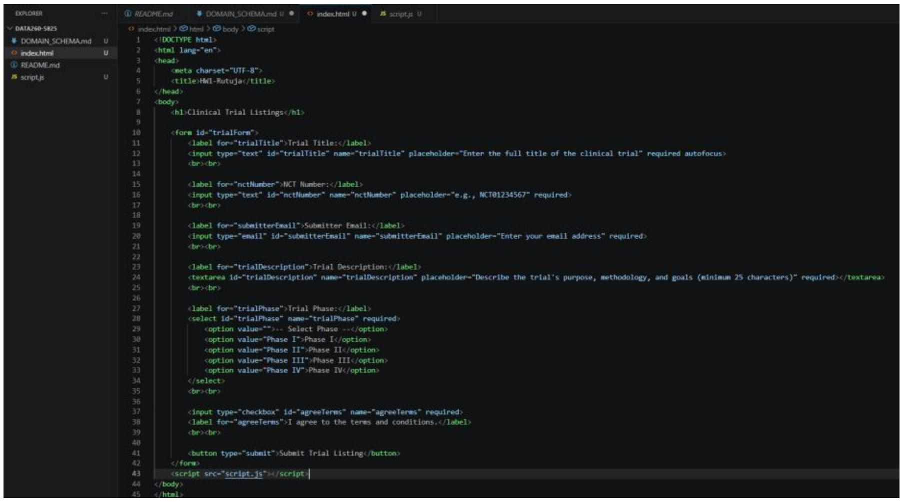
**OUTPUTS:**

PART I:

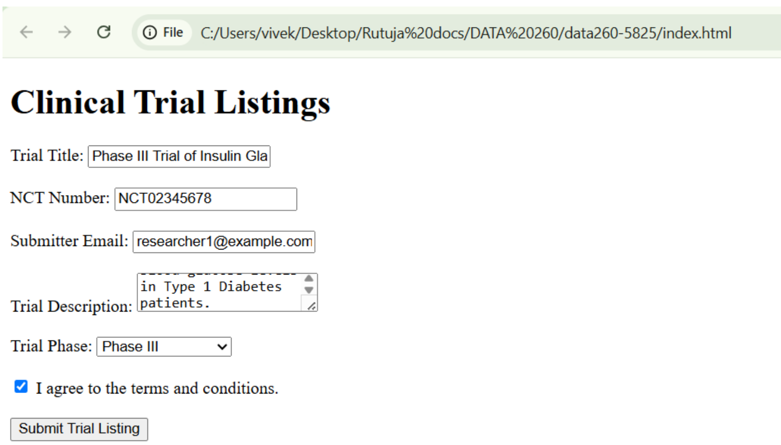
created DOMAIN_SCHEMA.md defining the clinical trial entity's fields (Trial Title, NCT Number, Submitter Email, Trial Description, Trial Phase, Terms Agreement) and the four Trial Phase category values (Phase I, Phase II, Phase III, Phase IV).

```
DOMAIN_SCHEMA.md > # Domain Schema- Clinical Trial Listings > ## Category Values — Trial Phase
# Domain Schema- Clinical Trial Listings

## Entity Fields

| Field | Type | Description |
|-------|------|-------------|
| Trial Title | Text (required) | Primary field — full name of the clinical trial |
| NCT Number | Text (required) | Secondary field — unique trial registry ID (format: NCT + 8 digits) |
| Submitter Email | Email (required) | Email of the person submitting the entry |
| Trial Description | Textarea (required) | Summary of the trial's purpose and methodology (must exceed 25 characters) |
| Trial Phase | Dropdown (required) | Category field, one of four fixed phase values |
| Terms Agreement | Checkbox (required) | Confirmation that the submitter agrees to terms and conditions |

## Category Values — Trial Phase

- Phase I
- Phase II
- Phase III
- Phase IV
```

Built an HTML form for clinical trial listings with fields for Trial Title, NCT Number, Submitter Email, Trial Description, Trial Phase, and terms agreement checkbox.



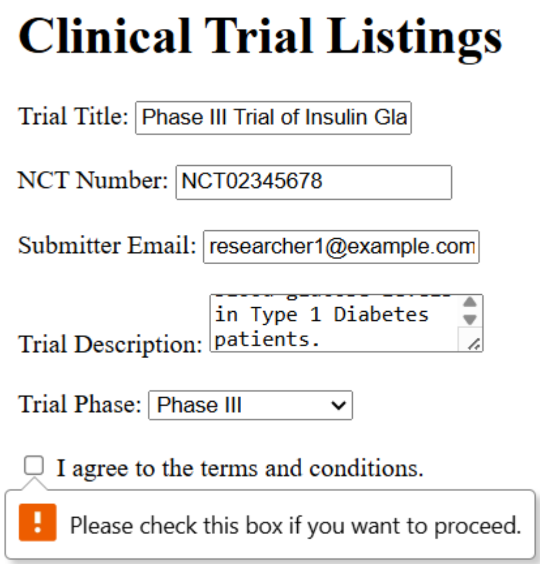
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>HW1-Rutuja</title>
</head>
<body>
    <h1>Clinical Trial Listings</h1>

    <form id="trialForm">
        <label for="trialTitle">Trial Title:</label>
        <input type="text" id="trialTitle" name="trialTitle" placeholder="Enter the full title of the clinical trial" required autofocus>
        <br><br>

        <label for="nctNumber">NCT Number:</label>
        <input type="text" id="nctNumber" name="nctNumber" placeholder="e.g., NCT01234567" required>
        <br><br>

        <label for="submitterEmail">Submitter Email:</label>
        <input type="email" id="submitterEmail" name="submitterEmail" placeholder="Enter your email address" required>
        <br><br>

        <label for="trialDescription">Trial Description:</label>
        <textarea id="trialDescription" name="trialDescription" placeholder="Describe the trial's purpose, methodology, and goals (minimum 25 characters)" required></textarea>
        <br><br>

        <label for="trialPhase">Trial Phase:</label>
        <select id="trialPhase" name="trialPhase" required>
            <option value="">-- Select Phase --</option>
            <option value="Phase I">Phase I</option>
            <option value="Phase II">Phase II</option>
            <option value="Phase III">Phase III</option>
            <option value="Phase IV">Phase IV</option>
        </select>
        <br><br>

        <input type="checkbox" id="agreeTerms" name="agreeTerms" required>
        <label for="agreeTerms">I agree to the terms and conditions.</label>
        <br><br>

        <button type="submit">Submit Trial Listing</button>
    </form>
    <script src="script.js"></script>
</body>
</html>
```




PART II:

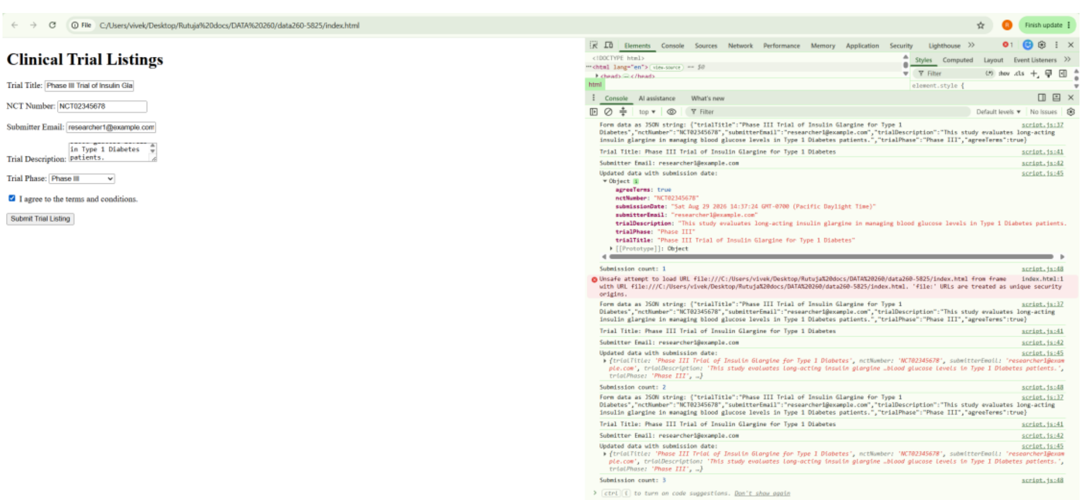
Javascript

Verified the alert pop ups if validation fails





Validated the console outputs: Console output showing JSON string, destructured values, updated object with submissionDate, and incrementing submission counts




**Docker file creation**

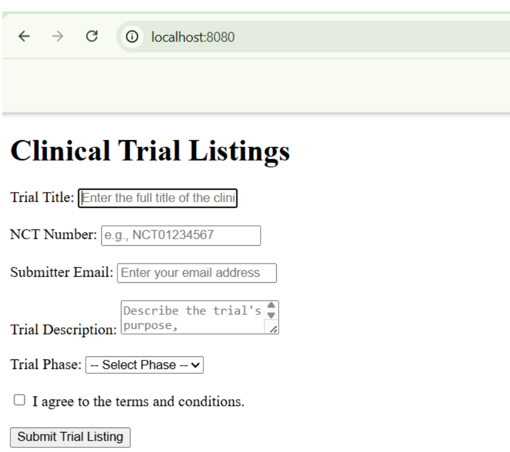
Created the docker image for clinical trial app: successfully build the Nginx-based image from the Dockerfile, confirming all layers were built and the image was tagged as clinical-trial-app:latest

```
C:\Users\vivek\Desktop\Rutuja docs\DATA 260\data260-5825>docker build -t clinical-trial-app .
[+] Building 3.9s (8/8) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                              0.0s
 => => transferring dockerfile: 344B                                                                              0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                                                   1.3s
 => [internal] load .dockerignore                                                                                 0.0s
 => => transferring context: 2B                                                                                   0.0s
 => [1/3] FROM docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffaeeb4a6d4b95a26fead017d5913  2.0s
 => => resolve docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffaeeb4a6d4b95a26fead017d5913  0.0s
 => => sha256:fb597529c91648bab2279939e01b745210875081852c73000aed882c000ff5de 20.50MB / 20.50MB                   1.6s
 => => sha256:8cdfe4f237782ae36780c4634b1457e8d6efa6f929fc3e46b0a2074ba54acd56 1.21kB / 1.21kB                     0.3s
 => => sha256:da4dea1b00afb53e163ae330a511ecd6978371d99794c80f439b83b609eb9d66 1.40kB / 1.40kB                     0.4s
 => => sha256:0d62d88506ba127db40e7c137e3a2dcf65e563709676828820b8c82277a6caeb 404B / 404B                         0.4s
 => => sha256:1ed8e39a7434c8975957bb78530f191c92eecbd9c7309966192ba276a7c0ac61 955B / 955B                         0.2s
 => => sha256:0ea93572787841bba2f3921ece8dc2db925c22232a1388c766881ebe3bd3f67d 628B / 628B                         0.2s
 => => sha256:d94291c262619626bc7a27b61b090d85365e85c61f92ca49b190a26930121299 1.90MB / 1.90MB                     0.6s
 => => sha256:55afa1ecc21d2bb5e5045f32dafee56272ffd89860bac26f6c32123439af26a4 3.85MB / 3.85MB                     0.7s
 => => extracting sha256:55afa1ecc21d2bb5e5045f32dafee56272ffd89860bac26f6c32123439af26a4                          0.1s
 => => extracting sha256:d94291c262619626bc7a27b61b090d85365e85c61f92ca49b190a26930121299                          0.1s
 => => extracting sha256:0ea93572787841bba2f3921ece8dc2db925c22232a1388c766881ebe3bd3f67d                          0.0s
 => => extracting sha256:1ed8e39a7434c8975957bb78530f191c92eecbd9c7309966192ba276a7c0ac61                          0.0s
 => => extracting sha256:0d62d88506ba127db40e7c137e3a2dcf65e563709676828820b8c82277a6caeb                          0.0s
 => => extracting sha256:8cdfe4f237782ae36780c4634b1457e8d6efa6f929fc3e46b0a2074ba54acd56                          0.0s
 => => extracting sha256:da4dea1b00afb53e163ae330a511ecd6978371d99794c80f439b83b609eb9d66                          0.0s
 => => extracting sha256:fb597529c91648bab2279939e01b745210875081852c73000aed882c000ff5de                          0.3s
 => [internal] load build context                                                                                 0.3s
 => => transferring context: 36.63kB                                                                              0.3s
 => [2/3] RUN rm -rf /usr/share/nginx/html/*                                                                      0.2s
 => [3/3] COPY . /usr/share/nginx/html/                                                                           0.0s
 => exporting to image                                                                                            0.2s
 => => exporting layers                                                                                           0.1s
 => => exporting manifest sha256:feaa1e44ca05cb5b15869bdefadccf3ba1c81b613b14ee16a474d4218908436e                  0.0s
 => => exporting config sha256:1dc8d60890c002ea7a805dfcbedc9a6094f72bd78e98ddf744a21601f530839e                    0.0s
 => => exporting attestation manifest sha256:dffdf03ec663a3975c57335862aa2658298f1af4dcdfed4adafaaff73aeea992      0.0s
 => => exporting manifest list sha256:00bd2738528866ac42d7749aa8e2e987795f7af464a151d893ba553cff42d21e             0.0s
 => => naming to docker.io/library/clinical-trial-app:latest                                                      0.0s
 => => unpacking to docker.io/library/clinical-trial-app:latest                                                   0.0s

C:\Users\vivek\Desktop\Rutuja docs\DATA 260\data260-5825>
```

Running the docker container and confirmed it

```
C:\Users\vivek\Desktop\Rutuja docs\DATA 260\data260-5825>docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS          PORTS                                   NAMES
6c8e582cfb66   clinical-trial-app   "/docker-entrypoint.…"   41 seconds ago   Up 40 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   clinical-trial-container
```

Application running on Localhost (Docker) - Clinical Trial Listings form successfully served via Docker/Nginx and accessed at http://localhost:8080, confirming successful local deployment.




**DEPLOYMENT:**

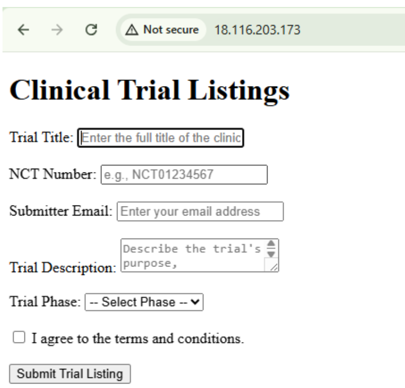
Docker Image Tagged and Pushed to Amazon ECR:

docker tag command successfully tagging the local image and

docker push command pushing all layers to the clinical-trial-app repository in Amazon ECR

```
C:\Users\vivek>docker tag clinical-trial-app:latest 093468662825.dkr.ecr.us-east-2.amazonaws.com/clinical-trial-app:late
st

C:\Users\vivek>docker push 093468662825.dkr.ecr.us-east-2.amazonaws.com/clinical-trial-app:latest
The push refers to repository [093468662825.dkr.ecr.us-east-2.amazonaws.com/clinical-trial-app]
c072f83ab0c3: Pushed
44136fa355b3: Pushed
d94291c26261: Pushed
55afa1ecc21d: Pushed
0ea935727878: Pushed
1ed8e39a7434: Pushed
0d62d88506ba: Pushed
8cdfe4f23778: Pushed
da4dea1b00af: Pushed
fb597529c916: Pushed
5acef8123c6a: Pushed
0938e7230cea: Pushed
latest: digest: sha256:00bd2738528866ac42d7749aa8e2e987795f7af464a151d893ba553cff42d21e size: 856
```

Application running on the public IP address from ECS




**Pipeline**: Planner, Reviewer, and Finalizer Output: Console transcript from running python agents_demo.py, showing the Planner agent's initial proposed tags and summary, the Reviewer agent's evaluation (which confirmed the Planner's output was already accurate), and the Finalizer's validated "Publish output" JSON containing exactly 3 tags and a summary within the 25-word limit.

1. Command used: python agents_demo.py
2. Console showing: Planner output, Reviewer output, Finalized output

```
(venv) C:\Users\vivek\Desktop\Rutuja docs\DATA 260\data260-5825>python agents_demo.py
=== Planner Agent ===
{"tags": ["Phase II Trial", "Metformin", "Type 2 Diabetes"], "summary": "Phase II trial evaluates Metformin's efficacy and safety in Type 2 Diabetes over 12 weeks, focusing on HbA1c reduction and adverse events
."}

=== Reviewer Agent ===
{"tags": ["Phase II Trial", "Metformin", "Type 2 Diabetes"], "summary": "Phase II trial evaluates Metformin's efficacy and safety in Type 2 Diabetes over 12 weeks, focusing on HbA1c reduction and adverse events
."}

=== Finalized Publish Output ===
{
  "tags": [
    "Phase II Trial",
    "Metformin",
    "Type 2 Diabetes"
  ],
  "summary": "Phase II trial evaluates Metformin's efficacy and safety in Type 2 Diabetes over 12 weeks, focusing on HbA1c reduction and adverse events."
}
```

Q1. The 3 final tags you obtained. –

"Phase II Trial", "Metformin", "Type 2 Diabetes”

Q2. The final summary (≤25 words).

Phase II trial evaluates Metformin's efficacy and safety in Type 2 Diabetes over 12 weeks, focusing on HbA1c reduction and adverse events

Q3. Did the Reviewer agent change anything? (yes/no +explanation)

No, As in my prompt design, I told the Reviewer it could either correct the Planner's output or keep it the same if it was already good and hence the Reviewer evaluated the Planner's tags ("Phase II Trial", "Metformin", "Type 2 Diabetes") and summary, and determined they were already accurate and well-formed, so it returned them unchanged. Given the prompt below:

*“def reviewer_agent(title, content, planner_output): prompt = f"""You are a Reviewer agent. You will review the Planner's proposed tags and summary for accuracy and quality, given the original title and content.”*

**Explanation of each step: Planner, Reviewer , Finalizer:**

Planner: Takes the input title as well as content and proposes an initial set of exactly 3 tags along with a draft summary which is less than 25 words. It returns the output in JSON format.

Reviewer: Takes the planner's proposed tags and summary, along with the original title/content, and checks them for accuracy and quality. It corrects them if required or keeps them unchanged if they're already good.

Finalizer: Takes the Reviewer's output, parses it, and makes sure there are exactly 3 tags and the summary is not too long that is not more than 25 words. It then produces the final validated "Publish output" in json format which is the final clean result.

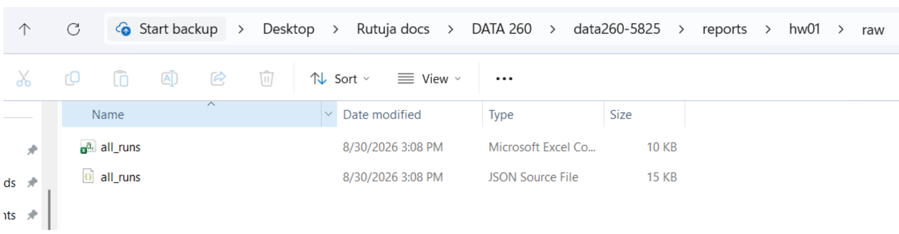
**Measuring Non-Determinism (Part 3):**

**Running the Agent 40 Times (20 runs at Temperature 0.7, 20 runs at Temperature 0.0):**

```
(venv) C:\Users\vivek\Desktop\Rutuja docs\DATA 260\data260-5825>python run_experiments.py
=== Starting 20 runs at temperature 0.7 ===
Running temp=0.7, run 1/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 24714.79ms
Running temp=0.7, run 2/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes', 'Phase II Trial'], Latency: 36030.62ms
Running temp=0.7, run 3/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 23105.28ms
Running temp=0.7, run 4/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes Management', 'Phase II Clinical Trial'], Latency: 28072.78ms
Running temp=0.7, run 5/20...
  Tags: ['Phase II Trial', 'Metformin', 'Type 2 Diabetes'], Latency: 39078.15ms
Running temp=0.7, run 6/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes', 'Phase II Trial'], Latency: 45094.62ms
Running temp=0.7, run 7/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes', 'Clinical Trial Phase II'], Latency: 54552.54ms
Running temp=0.7, run 8/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 27791.6ms
Running temp=0.7, run 9/20...
  Tags: ['Metformin Efficacy', 'Phase II Clinical Trial', 'Blood Glucose Management'], Latency: 28190.46ms
Running temp=0.7, run 10/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes', 'Phase II Trial'], Latency: 40165.82ms
Running temp=0.7, run 11/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes', 'Clinical Trial'], Latency: 40429.48ms
Running temp=0.7, run 12/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes', 'Clinical Trial Phase II'], Latency: 35527.82ms
Running temp=0.7, run 13/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes', 'Clinical Trial Phase II'], Latency: 44432.31ms
Running temp=0.7, run 14/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 25347.13ms
Running temp=0.7, run 15/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Blood Glucose Management'], Latency: 26358.0ms
Running temp=0.7, run 16/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes', 'Phase II Trial'], Latency: 45613.92ms
Running temp=0.7, run 17/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 35899.71ms
Running temp=0.7, run 18/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 29978.09ms
Running temp=0.7, run 19/20...
  Tags: ['Phase II Trial', 'Metformin Efficacy', 'Type 2 Diabetes'], Latency: 38643.75ms
Running temp=0.7, run 20/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes', 'Phase II Trial'], Latency: 42261.76ms
```

```
=== Starting 20 runs at temperature 0.0 ===
Running temp=0.0, run 1/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Metformin Safety'], Latency: 58469.29ms
Running temp=0.0, run 2/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes Management', 'Clinical Trial Safety'], Latency: 32533.61ms
Running temp=0.0, run 3/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 33518.69ms
Running temp=0.0, run 4/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes Management', 'Clinical Trial Safety'], Latency: 30482.58ms
Running temp=0.0, run 5/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 33516.66ms
Running temp=0.0, run 6/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes Management', 'Clinical Trial Safety'], Latency: 30744.5ms
Running temp=0.0, run 7/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 33531.37ms
Running temp=0.0, run 8/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes Management', 'Clinical Trial Safety'], Latency: 30452.11ms
Running temp=0.0, run 9/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 33493.14ms
Running temp=0.0, run 10/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes Management', 'Clinical Trial Safety'], Latency: 30390.39ms
Running temp=0.0, run 11/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 33478.53ms
Running temp=0.0, run 12/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes Management', 'Clinical Trial Safety'], Latency: 30410.33ms
Running temp=0.0, run 13/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 33543.15ms
Running temp=0.0, run 14/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes Management', 'Clinical Trial Safety'], Latency: 30411.64ms
Running temp=0.0, run 15/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 33511.32ms
Running temp=0.0, run 16/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes Management', 'Clinical Trial Safety'], Latency: 30398.42ms
Running temp=0.0, run 17/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 33525.61ms
Running temp=0.0, run 18/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes Management', 'Clinical Trial Safety'], Latency: 30374.8ms
Running temp=0.0, run 19/20...
  Tags: ['Phase II Clinical Trial', 'Metformin Efficacy', 'Type 2 Diabetes Management'], Latency: 33481.41ms
Running temp=0.0, run 20/20...
  Tags: ['Metformin Efficacy', 'Type 2 Diabetes Management', 'Clinical Trial Safety'], Latency: 30393.76ms

All 40 runs complete. Results saved to reports/hw01/raw/
```

Save the raw per-run results (tags + latency, all 40 runs) as JSON/CSV




Results Summary: Distinct Tags and Latency (Temp 0.7 vs Temp 0.0):

```
(venv) C:\Users\vivek\Desktop\Rutuja docs\DATA 260\data260-5825>python analyze_results.py

=== Temperature 0.7 ===
Distinct tag sets: 6
Tags in all 20 runs: []
Tags in exactly 1 run: ['Metformin', 'Clinical Trial']
Latency p50: 35965.17 ms
Latency p95: 45613.92 ms
Latency p99: 45613.92 ms

=== Temperature 0.0 ===
Distinct tag sets: 3
Tags in all 20 runs: ['Metformin Efficacy']
Tags in exactly 1 run: ['Metformin Safety']
Latency p50: 33006.07 ms
Latency p95: 33543.15 ms
Latency p99: 33543.15 ms

METRICS.md written to reports/hw01/METRICS.md
```

Q. Using your own numbers, explain what two users sending identical input might see,

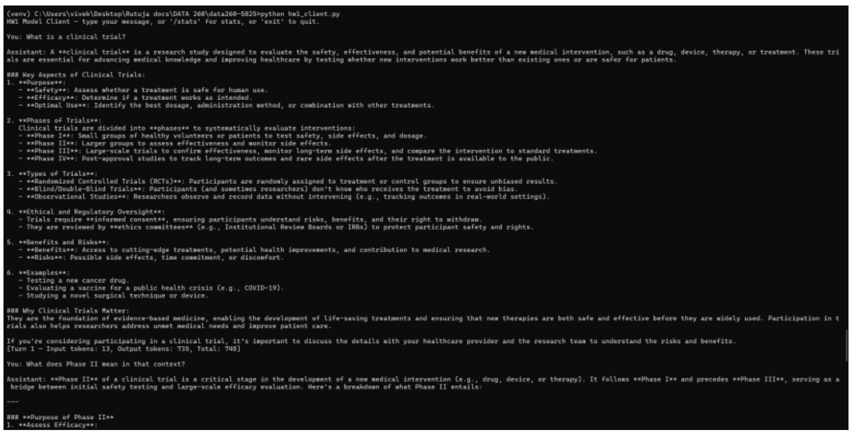
If two different people typed in the exact same clinical trial info, they might not get the exact same tags back. At the 0.7 temperature, we got 6 different combinations of tags across 20 tries, so two people would likely see different results. At the 0.0 temperature, one tag ("Metformin Efficacy") showed up almost every time, so two people might see the same thing sometimes but not always.

Q. and give one example where run-to-run variation is acceptable and one where it is not.

The variation is acceptable if this feature was just helping people browse and discover similar trials, it wouldn't matter if the tags change a little each time as people would still discover trials that are genuinely relevant to what they're looking for. And when the variation is random it is not acceptable if the same system was used for something official where exact same answer is expected every time, if it keeps changing it can lead to mistakes in official records and hence here more consistent setting of temperature 0.0 should be used instead of the random one.

**Model Client and Token Accounting (Part 4):**

**5-Turn Conversation with Token Tracking and /stats Checkpoints:**



```
(venv) C:\Users\vivek\Desktop\Rutuja docs\DATA 260\data260-5825>python hw1_client.py
HW1 Model Client - type your message, or '/stats' for stats, or 'exit' to quit.

You: What is a clinical trial?

Assistant: A **clinical trial** is a research study designed to evaluate the safety, effectiveness, and potential benefits of a new medical intervention, such as a drug, device, therapy, or treatment. These tri
als are essential for advancing medical knowledge and improving healthcare by testing whether new interventions work better than existing ones or are safer for patients.

### Key Aspects of Clinical Trials:
1. **Purpose**:
   - **Safety**: Assess whether a treatment is safe for human use.
   - **Efficacy**: Determine if a treatment works as intended.
   - **Optimal Use**: Identify the best dosage, administration method, or combination with other treatments.

2. **Phases of Trials**:
   Clinical trials are divided into **phases** to systematically evaluate interventions:
   - **Phase I**: Small groups of healthy volunteers or patients to test safety, side effects, and dosage.
   - **Phase II**: Larger groups to assess effectiveness and monitor side effects.
   - **Phase III**: Large-scale trials to confirm effectiveness, monitor long-term side effects, and compare the intervention to standard treatments.
   - **Phase IV**: Post-approval studies to track long-term outcomes and rare side effects after the treatment is available to the public.

3. **Types of Trials**:
   - **Randomized Controlled Trials (RCTs)**: Participants are randomly assigned to treatment or control groups to ensure unbiased results.
   - **Blind/Double-Blind Trials**: Participants (and sometimes researchers) don't know who receives the treatment to avoid bias.
   - **Observational Studies**: Researchers observe and record data without intervening (e.g., tracking outcomes in real-world settings).

4. **Ethical and Regulatory Oversight**:
   - Trials require **informed consent**, ensuring participants understand risks, benefits, and their right to withdraw.
   - They are reviewed by **ethics committees** (e.g., Institutional Review Boards or IRBs) to protect participant safety and rights.

5. **Benefits and Risks**:
   - **Benefits**: Access to cutting-edge treatments, potential health improvements, and contribution to medical research.
   - **Risks**: Possible side effects, time commitment, or discomfort.

6. **Examples**:
   - Testing a new cancer drug.
   - Evaluating a vaccine for a public health crisis (e.g., COVID-19).
   - Studying a novel surgical technique or device.

### Why Clinical Trials Matter:
They are the foundation of evidence-based medicine, enabling the development of life-saving treatments and ensuring that new therapies are both safe and effective before they are widely used. Participation in t
rials also helps researchers address unmet medical needs and improve patient care.

If you're considering participating in a clinical trial, it's important to discuss the details with your healthcare provider and the research team to understand the risks and benefits.
[Turn 1 - Input tokens: 13, Output tokens: 735, Total: 748]

You: What does Phase II mean in that context?

Assistant: **Phase II** of a clinical trial is a critical stage in the development of a new medical intervention (e.g., drug, device, or therapy). It follows **Phase I** and precedes **Phase III**, serving as a
 bridge between initial safety testing and large-scale efficacy evaluation. Here's a breakdown of what Phase II entails:

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### **Purpose of Phase II**
1. **Assess Efficacy**:
```
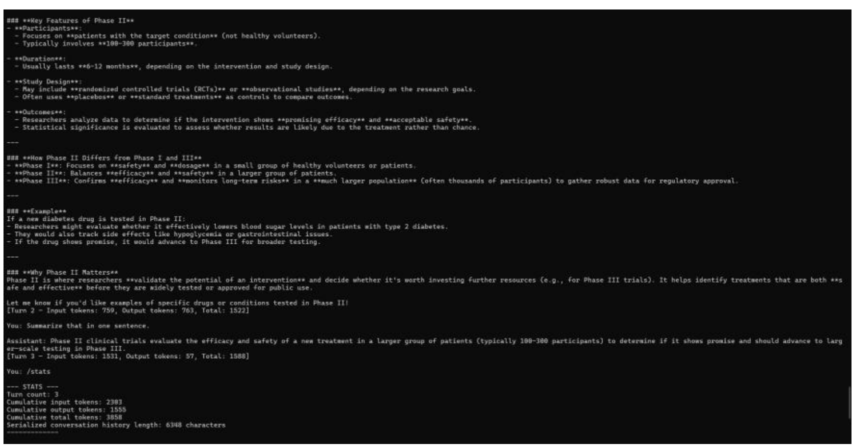

/stats output after turn 3



```
### **Key Features of Phase II**
- **Participants**:
  - Focuses on **patients with the target condition** (not healthy volunteers).
  - Typically involves **100-300 participants**.

- **Duration**:
  - Usually lasts **6-12 months**, depending on the intervention and study design.

- **Study Design**:
  - May include **randomized controlled trials (RCTs)** or **observational studies**, depending on the research goals.
  - Often uses **placebos** or **standard treatments** as controls to compare outcomes.

- **Outcomes**:
  - Researchers analyze data to determine if the intervention shows **promising efficacy** and **acceptable safety**.
  - Statistical significance is evaluated to assess whether results are likely due to the treatment rather than chance.

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### **How Phase II Differs from Phase I and III**
- **Phase I**: Focuses on **safety** and **dosage** in a small group of healthy volunteers or patients.
- **Phase II**: Balances **efficacy** and **safety** in a larger group of patients.
- **Phase III**: Confirms **efficacy** and **monitors long-term risks** in a **much larger population** (often thousands of participants) to gather robust data for regulatory approval.

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### **Example**
If a new diabetes drug is tested in Phase II:
- Researchers might evaluate whether it effectively lowers blood sugar levels in patients with type 2 diabetes.
- They would also track side effects like hypoglycemia or gastrointestinal issues.
- If the drug shows promise, it would advance to Phase III for broader testing.

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### **Why Phase II Matters**
Phase II is where researchers **validate the potential of an intervention** and decide whether it's worth investing further resources (e.g., for Phase III trials). It helps identify treatments that are both **s
afe and effective** before they are widely tested or approved for public use.

Let me know if you'd like examples of specific drugs or conditions tested in Phase II!
[Turn 2 - Input tokens: 759, Output tokens: 763, Total: 1522]

You: Summarize that in one sentence.

Assistant: Phase II clinical trials evaluate the efficacy and safety of a new treatment in a larger group of patients (typically 100-300 participants) to determine if it shows promise and should advance to larg
er-scale testing in Phase III.
[Turn 3 - Input tokens: 1531, Output tokens: 57, Total: 1588]

You: /stats

--- STATS ---
Turn count: 3
Cumulative input tokens: 2303
Cumulative output tokens: 1555
Cumulative total tokens: 3858
Serialized conversation history length: 6348 characters
=============
```

/stats output after turn 5

```
You: What is Metformin used for?

Assistant: **Metformin** is primarily used to **treat type 2 diabetes** by improving insulin sensitivity, reducing glucose production in the liver, and increasing glucose uptake in muscle cells. It is also some
times prescribed **off-label** for conditions like **polycystic ovary syndrome (PCOS)** to manage insulin resistance and **obesity** (though its role in weight loss is less well-established). It is generally co
nsidered a first-line therapy for type 2 diabetes due to its efficacy, affordability, and low risk of hypoglycemia.
[Turn 4 - Input tokens: 1595, Output tokens: 131, Total: 1726]

You: Thank you, that's all.

Assistant: You're welcome! If you have any more questions in the future, feel free to ask. Have a great day! 😊
[Turn 5 - Input tokens: 1732, Output tokens: 24, Total: 1756]

You: /stats

--- STATS ---
Turn count: 5
Cumulative input tokens: 5630
Cumulative output tokens: 1710
Cumulative total tokens: 7340
Serialized conversation history length: 7020 characters
-------------
```

```
You: exit

=== Session ended ===
Total turns: 5
Cumulative input tokens: 5630
Cumulative output tokens: 1710
Cumulative total tokens: 7340
```

**AGENT.md verification test output**

```
(venv) C:\Users\vivek\Desktop\Rutuja docs\DATA 260\data260-5825>python test_agent_review.py
=== Model Response ===
- The function is functionally correct for basic addition but lacks type hints for parameters and return value.
- Parameter names `a` and `b` are generic; more descriptive names would improve readability.
- The intermediate variable `result` is unnecessary; the function could directly return `a + b`.
- No error handling is included for non-numeric inputs, which could cause runtime errors.
- The function could benefit from a docstring to clarify its purpose and usage.
```

**Questions:**

1. why is prior conversation context resent with every turn?
   The AI has no memory. Everytime a message is sent the AI forgets everything the second the conversation ends. Hence to remember the conversation, the app must resend the full conversation every time just to remind it what's already been said.
2. How is a system prompt different from a user message?
   The system prompt is the fixed rule that is set at once for how the AI should behave whereas the user message is the actual question or task you ask it to do and it will change every time.
3. Why do input tokens grow over a conversation?

To remind AI the full conversation we have to resend it every time and hence it keeps getting bigger and bigger the more we talk. This is what my numbers showed, going from 13 tokens on turn 1 to over 1700 by turn 5.

4. What eventually limits that growth?
   Every AI model has a maximum size limit for how much text it can read at once, so eventually a long conversation would hit that limit and either break or need older messages trimmed out.

**Self-Check Verification Output:** Output of verify.py, confirming all required files are present, Python version is compatible, Ollama with the local model is available, and all 40 non-determinism runs completed successfully.

```
(venv) C:\Users\vivek\Desktop\Rutuja docs\DATA 260\data260-5825>python verify.py
{
  "required_files_present": true,
  "missing_files": [],
  "python_version": "3.11.9",
  "python_version_ok": true,
  "ollama_available": true,
  "total_runs_count": 40,
  "forty_runs_complete": true,
  "overall_status": "PASS"
}
```